

Justin Hatch

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Anthropic Partner Program member (Machine & Minds). Shipping production LLM systems at Modern Amenities; 1st @ QuackHacks II.

EDUCATION

University Of Oregon

B.S in Computer Science - Specialization in ML, AI & DS

Eugene, Oregon

Graduation: June 2026

- Dean's List | Department Honors | Oregon Blockchain Group | AI Club | QuackHacks II Winner

SKILLS

- **Languages:** Python (Pandas, NumPy, Scikit-learn), SQL, C, C++, TypeScript, React, Node.js
- **Backend & Infrastructure:** REST API, Linux, Microservices, CI/CD, Pytest, FastAPI, Pydantic, Postgres
- **DevOps & Tools:** Streamlit, Docker, Airflow, AWS, Git, MLflow
- **Machine Learning:** Feature Engineering, Predictive Modeling, XGBoost, PyTorch, HuggingFace Transformers, SHAP, Time Series Analysis, EDA, A/B Testing
- **ML Engineering:** Model Evaluation, Cross-Validation, Hyperparameter Tuning, Model Deployment, Inference Optimization, Experiment Tracking, Feature Pipelines
- **AI Engineering:** Anthropic & OpenAI API, LangChain, Multi-Agent Systems, RAG, Vector Databases, Tool Use, Structured Outputs, Prompt Caching, LLM-as-judge Evaluation
- **Certs:** [Data Camp - Associate Data Engineer in SQL](#)

PROJECTS

Churn Prediction + LLM Retention Email Pipeline - Python | Postgres | dbt | Dagster | XGBoost | MLflow | Claude | FastAPI | React

- Built churn-prediction to retention-email pipeline pairing **XGBoost + TreeSHAP with Anthropic Claude**, scoring **18,618 customers** nightly at **0.79 ROC-AUC** with **top SHAP** attributions persisted as structured prompt input.
- Implemented **tool-augmented** Claude email generation with structured outputs and **4 read-only Postgres tools**, grounding emails in live customer order and review history.
- Designed **LLM-as-judge eval** with success criteria anchors and enum-typed verdicts, shifting rubric distribution from 8.5–9.0 to 4.0–8.0 and surfacing generator bugs (wrong CTA intent, prohibited offers, tone-label mismatch).
- Orchestrated nightly **Dagster asset graph** (dbt build → XGBoost training → batch scoring + SHAP persistence → email generation → LLM-as-judge) over Postgres medallion, **enabling automated proactive outreach** for top 500 priority users.

Market Data Pipeline + AI News Synthesis - Python | FastAPI | Supabase | AWS | Dagster | Alpaca | Yahoo Finance | LangChain

- Designed a Postgres medallion schema (Bronze → Silver → Gold → AI) keyed by (ticker, interval, timestamp), enabling replayable analytics across **4.5M+ rows** and unblocking downstream ML/agent workloads.
- Built replayable ETL Dagster pipeline ingesting 230+ tickers across 1m–1d bars with conflict-aware upserts and historical backfills, achieving 100% idempotent reruns with zero duplicates across dozens of runs.
- Deployed a scheduled **LangChain** agent that **ingests news** and synthesizes market context, news, and financial reports into source-attributed summaries persisted to Postgres, powering downstream RAG workflows.

NBA Salary Efficiency Analysis - Machine Learning | Sports Analytics | Residual Analysis | Python

- Built an XGBoost regression model on 1,500+ player-seasons to quantify NBA market efficiency, achieving $R^2 = 0.372$ with the unexplained variance treated as mispricing signal, surfacing the top 10 over/underpaid contracts league-wide.
- Tuned **700** hyperparameter configs with 4-fold CV and performed **residual analysis** to surface over/underpaid signals as **decision-ready insights**.
- Engineered features (OBPM, DBPM, VORP, position encoding, log transforms) to capture pay–performance relationships.

Evergreen Capital AI-Governed On-Chain Prediction Fund - Python | FastAPI | Gemini | TypeScript

- Architected a 5-agent Gemini research desk (Quant, Macro, Skeptic, Data Miner, Trader) where agents independently research, debate via structured cross-examination, and produce weighted consensus votes to trigger autonomous Polymarket trades.
- Built the FastAPI backend handling agent debate orchestration, weighted vote aggregation, and signal handoff to the on-chain execution layer, enabling autonomous trade triggers without human-in-the-loop.
- Built full-stack MVP in 24 hours with 2 teammates: **1st Place Solana Track, 2nd Place Polymarket Track at QuackHacks II.**

EXPERIENCE

Modern Amenities

Coburg, Oregon

AI Engineering Intern - Python | FastAPI | Claude | Prompt Caching | Prompt Engineering & Eval

March 2026 - Present

- Built FastAPI sales chatbot using Claude API qualifying prospects across **dozens of machine SKUs** in a 3D/AR demo serving **900+ app users**; used tool-use API to emit structured calls so frontend renders deterministic CTAs.
- Hardened system prompt against jailbreak vectors via delimited untrusted input and output scanning, achieving **100% block rate with zero prompt leaks**.
- Cut inference cost by implementing prompt caching with ephemeral **cache_control** on the static prefix and a 24h token budget, reducing input token costs by **upwards of 90%** on cached requests.

Oregon Blockchain Group - Deployment | CI/CD | Git | GitHub | AWS S3 | AWS App Runner

Eugene, Oregon

Lead Software Engineer (promoted from Software Developer)

October 2025 – June 2026

- Led 2 engineering projects, **Crop Share** (farm to store supply-chain provenance) and **TrialWeave** (GLP-1 cohort RWE analytics) translating stakeholder needs into scoped specs and acceptance criteria, shipping across a team of **3-5 developers**.
- **Drove 30%+ membership growth in under a year** by serving on the recruitment board, pitching in core CS classes, and interviewing potential candidates.